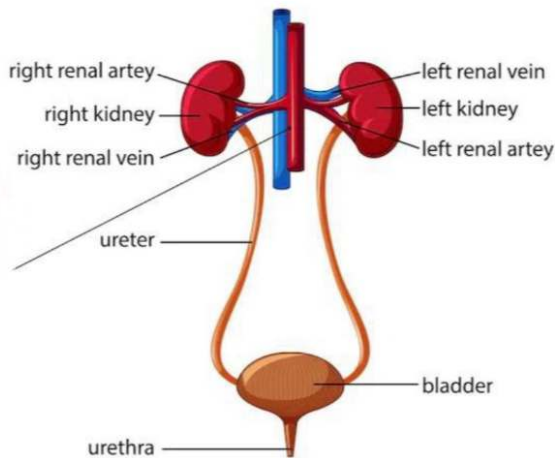


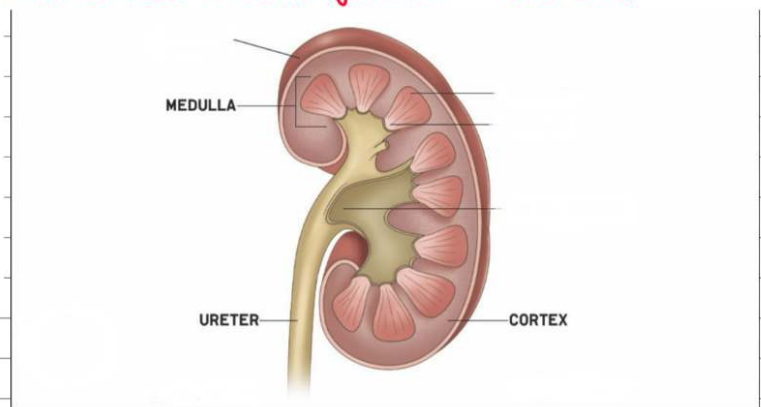
Excretion

Carbon dioxide is excreted through the lungs
The kidneys excrete urea and excess water and ions

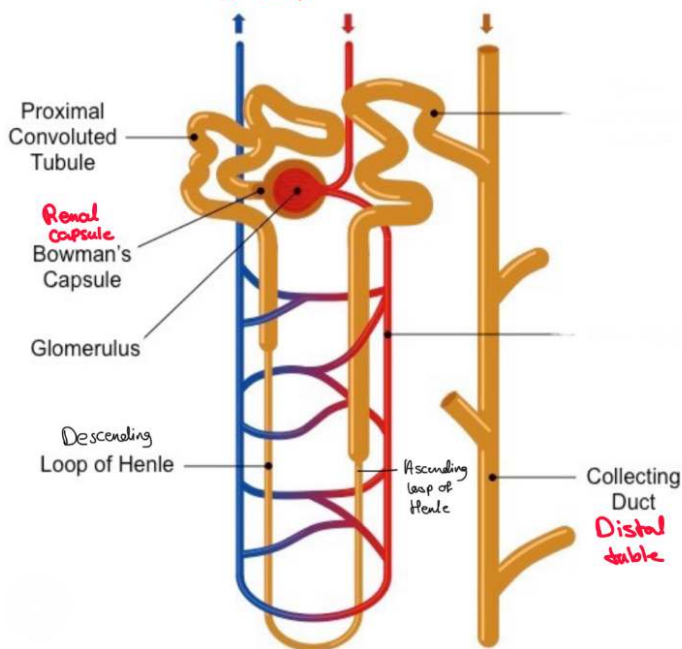
Excretion: The removal of waste products of metabolism and substances in excess of requirement. It also includes the removal of drugs or other foreign substances taken into the alimentary canal and absorbed by the blood



Kidneys: Remove urea and excess water from the body and reabsorb glucose and some salts back into the body
Ureter: A tube that delivers urine from kidneys to bladder
Bladder: A bag that stores urine
Urethra: Allows urine to get out of the body



Nephron



Glomerulus: Where filtration of the blood from water, glucose, urea and ions, happens to the renal capsule.

Proximal convoluted tubule: Reabsorbs glucose back into the blood by active transport.

Reabsorption: The filtration from the glomerulus collects the renal capsule and passes down the renal tube. The capillaries that surround the tubes absorb these substances back into the blood.

Descending loop of henle reabsorbs water because it is permeable to water but not salts. Water moves by osmosis

Ascending loop of henle reabsorbs salts because it is permeable to salts but not water. Salts move by diffusion.

Distal tubule: Reabsorbs remaining water by osmosis into the blood and stores urin until it goes to the ureter.

Function of the liver: Converts excess amino acids into proteins

* Urea is formed in the liver from excess amino acids

→ **Deamination:** The removal of the nitrogen-containing part of amino acids to form urea

Digested amino acids are absorbed into the blood and taken to the liver through the hepatic portal vein. Amino acids that are needed are released back into the circulation. Excess amino acids are deaminated to carbohydrates and ammonia by removing the nitrogen group. Ammonia is converted to urea. Urea is taken by the blood to the kidneys

Excretion is very important because it removes the toxic substances (urea) from the body.